

Email Corpus vs. Categorization Schema — Fit & Gap Analysis

Fitting the UPTET-2026 email helpline corpus in [data/](#) against the 17-category [CATEGORIZATION_SCHEMA](#) (see [taxonomy.md](#)), and identifying what the schema is missing.

Status: applied 2026-08-14. All 14 proposed sub-categories and the duplicate fix are now live in [geminiService.js](#). The schema went 184 → 196 sub-categories and coverage rose **79.5% → 82.4%**. Percentages in the *Result* section below are the **pre-change** baseline, kept as the record of why each change was made; the post-change figures are in [Outcome](#).

Source

Files	9 per-day sheets + 1 summary sheet, exported from Email Issues Uptet updated.xlsx
Period	27 Mar 2026 – 4 Apr 2026 (detail sheets); summary covers 27 Mar – 30 Apr
Rows with issue text	1,895 (of ~2,010 raw; the rest are blank spreadsheet padding)
Languages	English, Devanagari Hindi, Hinglish — frequently mixed within one email
Pre-existing labels	189 distinct free-text human categories, 313 rows blank, replied done leaking in from the status column

The summary sheet separately reports 13,041 emails received over the full window at a 91.32% reply rate, so **the 1,895 detail rows are roughly a 15% sample** — early-window, application-phase traffic.

Method and its limits

Rule-based classifier (~90 regex patterns across all three scripts), first-match-wins, most-specific-first, validated so every rule targets a real schema entry. Script: [scratchpad/classify.js](#).

This is a **coverage estimate, not a labelled gold set**. There is no hand-labelled ground truth to score against, patterns are recall-oriented, and broad buckets ([B.Ed / D.El.Ed Qualification Entry, Final Year Appearing Candidate Entry](#)) over-capture. Treat category-level percentages as reliable to a few points and individual sub-category counts as indicative. The gap counts in the proposals section were measured independently of the classifier and are the more trustworthy numbers.

Result: 79.5% of the corpus fits the existing schema

1,507 of 1,895 rows matched; 388 did not.

Category	Rows	Share
Educational Qualifications	374	19.7%

Category	Rows	Share
Payment & Fee	299	15.8%
Identity Verification	201	10.6%
OTR Completion & Preview	129	6.8%
Portal Access & Registration	120	6.3%
Exam Application & Eligibility	115	6.1%
<i>(triage disposition — not an issue)</i>	62	3.3%
Address & Personal Details	62	3.3%
Category & Reservation	60	3.2%
Login & Account Access	29	1.5%
Amendment & Post-Submission	27	1.4%
Uploads & Documents	16	0.8%
OTP, Password & CAPTCHA	10	0.5%
Exam Information	3	0.2%
Unmatched	388	20.5%

Top sub-categories by volume:

Rows	Sub-category
207	Educational Qualifications → Final Year Appearing Candidate Entry
110	Payment & Fee → Money Debited but Application Incomplete
98	Payment & Fee → Duplicate Payment Refund Query
96	OTR Completion & Preview → How to Edit Saved OTR Data
88	Identity Verification → Live Photo / Face Match Failure
85	Educational Qualifications → B.Ed / D.El.Ed Qualification Entry
77	Payment & Fee → Payment Pending / Processing Status
69	Portal Access & Registration → Registration Form Submission Error
63	Identity Verification → Name / DOB Mismatch Across Documents
49	Exam Application & Eligibility → Practising Government Teacher Details Entry

The headline: three themes — appearing/final-year candidates, payment reconciliation, and identity verification — account for roughly half of all traffic.

What the 388 unmatched rows are

Class	Rows	Action
Genuinely missing sub-categories	~52	Add — see proposals below
Too vague to classify (≤ 4 words: Payment , OTR ISSUE , Correction , Unable to proceed)	72	Not a taxonomy problem — an intake problem
Agent's own reply text pasted into the Issue column	4	Data quality
Follow-up chasers on an existing ticket	3	Needs a Follow-Up disposition, not a category
Long tail of one-off queries	~257	Expected residue; no action

Dormant schema — do not delete

Four categories scored **zero hits**, and 126 of 184 sub-categories were never used:

- Admit Card & Certificate
- Scribe & Compensatory Time
- Result & Merit List
- General Enquiry

This is **not** evidence they're dead. The corpus is the application window (late March / early April 2026); admit-card traffic arrives near the exam, result traffic after it, and scribe requests cluster around the admit-card release. Re-run this analysis on a post-exam corpus before touching them.

General Enquiry scoring zero is different, and is a naming problem rather than a demand problem — its sub-categories (**Call Back Request**, **Unrelated / Wrong Call**, **Repeated Call / Follow-up**) are all phrased for voice calls and none can match an email. See *Structural recommendations*.

Additions applied

The schema's own generator prompt requires every category to trace back to **at least 3 source items** (`geminiService.js:528-531`). Applying that bar to the corpus, these 14 clusters qualified — all are now in the schema. Counts below are the independent regex matches that justified each one; the *Outcome* section reports what each actually captured once wired into the classifier.

Hits	Proposed sub-category	Parent category	Rationale
33	Name Prefix Mismatch (KM / Kumari)	Identity Verification	Marksheet reads KM AASHA , Aadhaar reads ASHA . Distinct, highly repetitive failure mode; currently absorbed into the generic Name / DOB Mismatch Across Documents
23	Appearing Option Missing in Qualification Dropdown	Educational Qualifications	<i>A portal defect</i> , not an eligibility question. Currently collapses into Final Year Appearing Candidate Entry , hiding a fixable bug inside a query bucket

Hits	Proposed sub-category	Parent category	Rationale
20	Surname Changed After Marriage — eKYC Mismatch	Identity Verification	Aadhaar updated to husband's surname, board records not. Same argument as above
19	Wrong Exam Level Selected (Primary vs Junior)	Amendment & Post-Submission	Candidate clicked Junior instead of Primary; several report no back/previous option. Distinct from Subject / Paper Change After Submission
19	Apply / OTR Link Not Visible on Portal	Portal Access & Registration	"Apply Now ka option nhi aa rha" — the option is absent, not the page broken. Portal Not Loading / Technical Error is the wrong home
11	Wrong Person's Document Uploaded	Uploads & Documents	Candidate uploaded another individual's Aadhaar; needs a distinct remediation path
9	Board Roll Number Digit-Length Rejected	Educational Qualifications	Form demands ≥ 5 digits; older board roll numbers are 4. A validation bug with a clean signature
8	Out-of-State / NIOS Qualification Not Listed	Educational Qualifications	NIOS D.El.Ed and other-state D.El.Ed have no selectable option. Broader than Board / University Not in Dropdown — it's an eligibility question too
7	Handwritten Declaration Language Requirement	Uploads & Documents	Hindi vs English. Existing Handwritten Declaration Upload Issue covers upload mechanics only
5	In-Service Teacher Without Graduation — Eligibility	Exam Application & Eligibility	Pre-1996/2004 appointees holding only Intermediate + training. Practising Government Teacher Details Entry covers data entry, not eligibility
3	Graduation Field Mandatory but Not Applicable	Educational Qualifications	B.El.Ed and similar integrated-course holders cannot leave the field blank
3	Wrong ID Type Used at Registration	Identity Verification	Registered with PAN/DL by mistake, wants to switch to Aadhaar. Identity Proof Selection Query covers choosing, not changing
3	Marital Status Field Query	Address & Personal Details	Not represented anywhere in the current schema
3	Form Cancellation / Withdrawal Request	Amendment & Post-Submission	"Please reject or cancel this form" — no destination today

Below the evidence bar, recorded but **not** recommended yet: Second Language Selection Error (2), Email Domain Not Accepted (1), Refund Timeline Query (1).

One split worth making  applied

Final Year Appearing Candidate Entry was carrying 207 rows — 11% of the entire corpus — and doing two different jobs. Splitting the portal defect out into **Appearing Option Missing in Dropdown** leaves the eligibility query bucket clean and makes the defect countable for the engineering team. Post-change the eligibility bucket sits at 182 rows, with 26 rows now correctly filed as a defect.

Outcome after applying the changes

Schema: **184** → **196** sub-categories (14 added, 2 duplicates removed). Coverage: **79.5%** → **82.4%**; unmatched rows **388** → **334**.

Rows captured	New sub-category
33	Name Prefix Mismatch (KM / Kumari)
26	Appearing Option Missing in Dropdown
21	Post-Marriage Surname Mismatch in eKYC
20	Apply / OTR Link Not Visible on Portal
19	Wrong Exam Level Selected (Primary vs Junior)
13	Wrong Person's Document Uploaded
12	Out-of-State / NIOS Qualification Not Listed
9	Board Roll Number Digit Length Rejected
7	Handwritten Declaration Language Requirement
4	Graduation Field Mandatory but Not Applicable
4	In-Service Teacher Without Graduation Eligibility
3	Wrong ID Type Used at Registration
3	Form Cancellation / Withdrawal Request
3	Marital Status Field Query
177	total — 9.3% of the corpus

Every one clears the ≥ 3 bar against live data, not just against the exploratory patterns. Note that 177 captured but only 54 fewer unmatched: the remaining ~123 were pulled *out of over-broad legacy buckets* where they had been miscounted — which is the more valuable half of the change.

Category shares after the revision:

Category	Before	After
Educational Qualifications	19.7%	20.3%
Payment & Fee	15.8%	15.4%
Identity Verification	10.6%	12.2%

Category	Before	After
Portal Access & Registration	6.3%	7.1%
OTR Completion & Preview	6.8%	6.1%
Exam Application & Eligibility	6.1%	5.8%
Amendment & Post-Submission	1.4%	2.5%
Uploads & Documents	0.8%	1.8%

Still open

Two of the three structural recommendations below are **not** implemented — they are larger changes than a schema edit:

- The **disposition axis** needs a new field plus worker logic, not just schema entries.
- **Channel-neutral renaming** needs a retro-remap of historical records via `/generalise-categories`.

Both remain the right next steps.

Structural recommendations

1. Dispositions belong on their own axis

62 rows (3.3%) are not issues at all — they're triage outcomes. The summary sheet independently accounts for **1,132 non-actionable emails (8.7% of 13,041)**, described as Duplicate, Blank, Bank Revert, and Thank You.

Filing `Blank Email` under a category would worsen the MECE problems the schema already has. Your codebase already handles this informally — `Call too Short` and `Audio Unclear` are dispositions that bypass `CATEGORIZATION_SCHEMA` entirely. Make it explicit:

```
disposition: Actionable | Duplicate | Blank | Courtesy/Thank You
              | Auto-Reply | Bank Notification | Follow-Up | Unclear
```

Evaluated first; only `Actionable` items get a `category` / `sub_category`. This also gives the 72 too-vague rows a home (`Unclear`) instead of forcing a bad guess.

2. The schema is call-shaped and email is now in scope

`Call Back Request, Unrelated` / `Wrong Call, Repeated Call` / `Follow-up, Helpline Timing Query` cannot match an email — which is exactly why `General Enquiry` scored zero. Either rename to channel-neutral wording (`Callback Request` → `Contact Request, Unrelated` / `Wrong Call` → `Out-of-Scope Contact`) and add a `channel: call | email` field, or accept that one schema cannot serve both. Renaming is the cheaper path and the retro-remap machinery in `/generalise-categories` already exists to migrate historical records.

3. Fix the two duplicate sub-categories first ☒ applied

Password Forgotten / Reset and **OTR ID Forgotten** appeared under both #3 and #11, so any classifier — Gemini or otherwise — split those calls arbitrarily between two parents. Resolved before the additions went in: #3 now owns credential challenges (password, OTP, CAPTCHA) and #11 owns account access (login, lockouts, identifier recovery, sessions). Details in the [change log](#).

Data quality notes for the intake process

These affect what any classifier can achieve, independent of taxonomy design:

- **72 rows (3.8%) are too vague to classify** — single words like **Payment**, **OTR**, **Correction**. If the Issue field were captured from the email subject *and* first body paragraph rather than a hand-typed gist, most would become classifiable.
- **The human category column is unusable as training data** — 189 distinct labels for 1,895 rows, with **otr issue/otr/registation issue/registration failed issue** as separate entries, **eligibility/eligibility**, and **ediiting/editing** typos. 313 rows blank; 36 contain the status value **replied done**.
- **Sub-category was captured on only 521 of 1,895 rows** (27%).
- **Response rate degrades under load** — days ≥ 400 emails average 89.78% reply rate vs 94.78% on lighter days; the 25 Apr spike (2,400 emails, 9× the 260 median) fell to 83.42%. A staffing signal, not a taxonomy one.

Reproducing

```
node scripts/taxonomy-analysis/normalize.js data # 9 sheets ->
data/_normalized.json
node scripts/taxonomy-analysis/classify.js data # -> data/_classified.json +
coverage report
```

normalize.js merges the per-day sheets (headers differ on every sheet) into one array via an RFC4180 parser that handles the embedded newlines in the Issue column. **classify.js** imports **CATEGORIZATION_SCHEMA** directly from **backend/src/services/geminiService.js**, so it fails loudly if a rule ever targets a sub-category that no longer exists.